How Much of Genomic Language Model Variant-Effect Prediction Is Base Composition?

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Abstract

Zero-shot variant-effect prediction (VEP) with genomic language models (gLMs) scores a variant by the log-likelihood ratio LLR $= \log p(\text{alt}) - \log p(\text{ref})$. This quantity mixes learned regulatory function with a purely local signal, namely how mutationally expected a substitution is given its base composition and sequence context. We isolate and quantify this composition/mutation-expectedness axis as a confound on real clinical and functional VEP benchmarks, across eight gLMs and four benchmark families, using only context-free features that require no learning. A composition-only classifier reaches AUROC 0.720 on noncoding ClinVar and recovers 45–53% of the above-chance AUROC of every leading model, including Evo2-40B, GPN-MSA, CADD and phyloP; the result is unchanged when coding-overlapping variants are removed. On carefully composition-matched benchmarks (TraitGym) the same models drop to roughly a fifth to a quarter, so matching removes half of the confound but not all of it. The confound is strongly model dependent: smaller single-sequence gLMs (Nucleotide Transformer, Caduceus, HyenaDNA) do not exceed the composition baseline on any benchmark, while alignment-based (GPN-MSA) and large autoregressive (Evo2-40B) models retain signal that survives composition matching and concentrates on TF-motif-disrupting variants (odds ratio up to 5.4). Within saturation-mutagenesis screens with fixed background the leakage is negligible, which localises the confound to across-set ascertainment rather than local context. We provide a model-agnostic post-hoc pentanucleotide calibration that halves the mutation-rate coupling of the most affected model at negligible AUROC cost, and release it as a drop-in evaluation transform.

Keywords: genomic language models, variant effect prediction, benchmark confounds, base composition, mutation rate, calibration

1. Introduction

Genomic language models (gLMs) are increasingly used as zero-shot predictors of noncoding variant effect: a pretrained model scores a single-nucleotide variant (SNV) by the log-likelihood ratio LLR $= \log p(\text{alt}) - \log p(\text{ref})$, ranked against clinical or functional labels (Benegas et al., 2025a; Brixi et al., 2026). Aggregate AUROCs on noncoding ClinVar routinely exceed 0.95, read as evidence that the models have learned regulatory grammar.

Recent work complicates that reading. On synthetic promoters, gLM compositional sensitivity tracks AT content almost perfectly and a small position-aware model beats billion-parameter gLMs (Anonymous, 2026). Stratifying ClinVar by region type collapses Evo2's aggregate noncoding AUROC, because pathogenic variants concentrate in easy regions (Lu et al., 2025). Cyclic permutation of flanking context collapses Evo2 tRNA sensitivity, so predictions ride on irrelevant context (Mathur and Sachidanandam, 2026). Frozen gLM representations give little advantage over one-hot encodings on regulatory tasks (Tang et al., 2025), and the usable information concentrates in embedding rather than transformer layers (Rochkoulets et al., 2026).

These identify positional-grammar, region-type and context-artifact confounds. We isolate a distinct axis on real VEP: local *base composition* and *mutational expectedness*. The mechanism is direct. A model's likelihood partly encodes how expected a substitution is under neutral evolution:

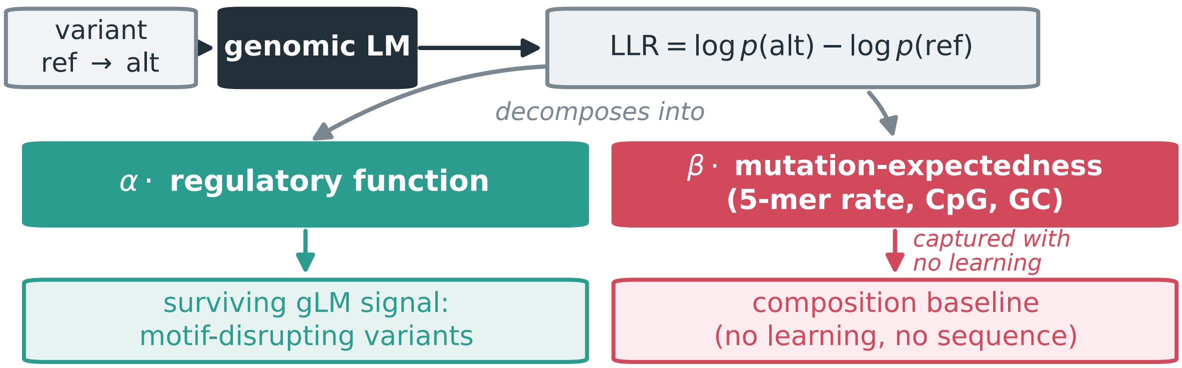

Figure 1: **The confound in one picture.** A zero-shot gLM scores a variant by its log-likelihood ratio, which mixes a regulatory-function term with a mutation-expectedness term set by local base composition. The second term needs no learning: a context-free baseline recovers about half of gLM AUROC on unmatched benchmarks, a post-hoc pentanucleotide calibration removes it, and the signal that survives concentrates on motif-disrupting variants.

CpG transitions mutate an order of magnitude faster than transversions, and mutation subtype couples to CpG and GC content (Carlson et al., 2018). Because pathogenic variants are ascertained non-randomly across mutation types, this component of LLR correlates with labels for reasons unrelated to function, which is why CADD and GPN-MSA prefer gnomAD-common over ClinVar-benign controls (Rentzsch et al., 2019; Benegas et al., 2025a). GPN-Star found raw gLM scores track pentanucleotide mutation rate at $\rho = 0.31$–0.34 and calibrated a single model at training time (Benegas et al., 2025c).

Contributions. We turn that observation into a benchmark-wide, cross-model, post-hoc audit. (1) We show a context-free composition baseline recovers about half of the above-chance AUROC of every leading model on unmatched ClinVar, and about a fifth on composition-matched TraitGym, which quantifies how much aggregate-leaderboard performance is composition-attributable (§4.2). (2) We show the confound is model-heterogeneous: smaller single-sequence gLMs are indistinguishable from the baseline, whereas alignment-based and large autoregressive models keep signal that survives composition matching (§4.1, §4.3). (3) We localise the confound to across-set ascertainment rather than local context using within-element saturation mutagenesis (§4.4). (4) We generalise pentanucleotide calibration into a model-agnostic post-hoc transform and release it (§4.5). (5) We map where genuine gLM signal survives, namely TF-motif-disrupting variants (§4.6). Our framing follows the confound-audit genre of Gupta et al. (2025): name a hidden confound, build a clean evaluation, measure its effect across a model class, and localise where real signal remains.

2. The composition and mutation-expectedness confound

We define two entangled axes, both computed with no learning, for an SNV at position p (Figure 1).

C1, local composition change. For window half-width w, let s_{ref} and s_{alt} be the $\pm w$ sequences. We take Δ_k, the L_1 distance between the k-mer frequency spectra of s_{ref} and s_{alt} for $k \in \{1, 2, 3\}$, and $\Delta\text{GC} = |\text{GC}(s_{\text{alt}}) - \text{GC}(s_{\text{ref}})|$, sweeping $w \in \{11, 21, 51\}$ bp.

C2, mutational expectedness. The pentanucleotide neutral mutation rate for the specific substitution (Carlson et al., 2018), the CpG status of the reference and alternate sites, the transition/transversion class, and the flanking GC background.

All features are strand-invariant. Under the model $\text{LLR} \approx \alpha \cdot (\text{function}) + \beta \cdot (\text{expectedness}) + \varepsilon$, the confound is the β term, which tracks labels through ascertainment rather than function; matching and calibration target β, leaving α as the gLM's genuine contribution beyond a mutation-rate lookup.

3. Experimental setup

Benchmarks. *ClinVar noncoding* (153,508 SNVs, 25,571 pathogenic): ≥2-star pathogenic/likely-pathogenic versus benign/likely-benign SNVs overlapping an ENCODE candidate cis-regulatory element (cCRE) (ENCODE Project Consortium, 2020); region-heterogeneous and unmatched, so the confound should be largest. *TraitGym* Mendelian (3,380; 338 positive) and complex (11,400; 1,140) (Benegas et al., 2025b): fine-mapped causal regulatory variants matched on chromosome, consequence, distance to transcription start site (TSS), and for complex traits also minor allele frequency (MAF) and linkage disequilibrium (LD), so the confound should be smallest. *Kircher* saturation massively-parallel reporter assay (MPRA) (Kircher et al., 2019): 39,170 SNVs across 21 elements with a continuous $\log_2$ expression effect and fixed within-element background. *Genomics Long-Range Benchmark (GLRB) causal expression QTL* (InstaDeep et al., 2024): 8,850 held-out test SNVs where zero-shot DNA LMs are known to be near chance.

Models. We use authors' released per-variant scores throughout, so model quality is not confounded by our inference. TraitGym supplies eight gLMs (GPN-MSA, an alignment-based multiple-sequence-alignment model; GPN, its single-sequence counterpart; Evo2-7B/40B; Nucleotide Transformer; Caduceus; HyenaDNA; SpeciesLM) plus CADD, phyloP and phastCons. On ClinVar we use precomputed Evo2-7B/40B and Nucleotide Transformer scores, a genome-wide GPN-MSA tabix lookup, CADD, phyloP and SpliceAI. GPN-MSA, the alignment-based anchor, appears on every benchmark. Our composition baseline is a histogram gradient-boosted-tree classifier (250 trees, learning rate 0.05) on the 19 C1/C2 features only, with no sequence context; a small one-hot CNN on ±100 bp windows serves as a learned-from-scratch reference.

Protocol. We report the area under the ROC curve (AUROC) and the precision-recall curve (AUPRC), oriented so higher scores mark the positive class. Binary benchmarks use leave-one-chromosome-out cross-validation with sample-size-weighted averaging (Benegas et al., 2025b), partial Spearman correlations controlling for consequence or region, the paired DeLong test with Benjamini–Hochberg control across model×benchmark comparisons, and percentile bootstrap intervals. The composition classifier uses the same folds, so no variant informs its own fold.

4. Results

4.1. E1: gLM scores encode the confound, model-dependently

Figure 2(a) reports the partial Spearman correlation between |LLR| and the pentanucleotide mutation rate, controlling for consequence. GPN-MSA is the most strongly coupled model at $\rho = -0.33$ on ClinVar and -0.16 to -0.20 on TraitGym, consistent with an alignment model that tracks

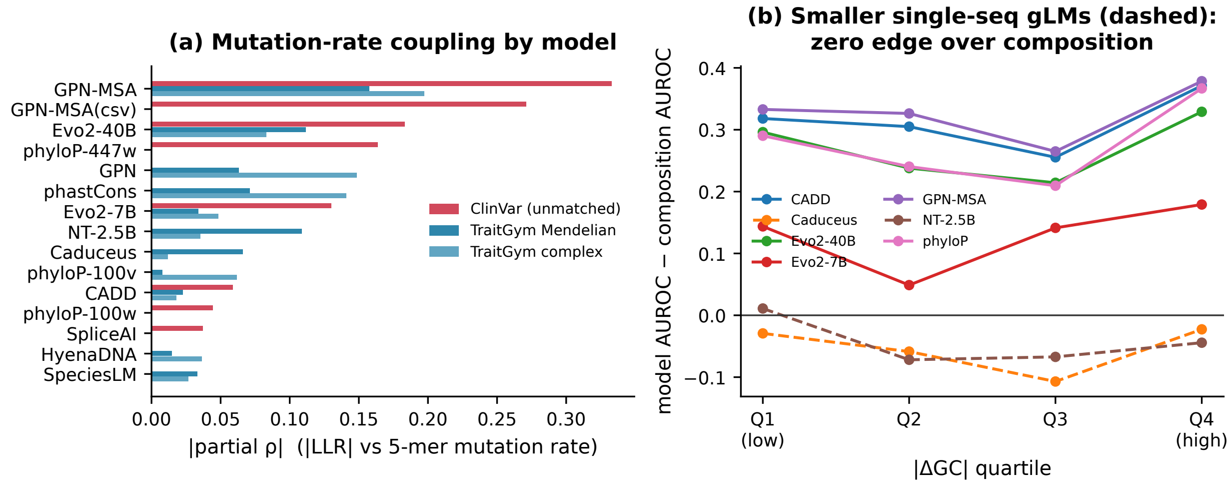

Figure 2: **Confound structure.** (a) Partial $|\rho|$ between |LLR| and the 5-mer mutation rate per model and benchmark. GPN-MSA is most coupled. (b) Collapse curve on TraitGym Mendelian: model AUROC minus composition AUROC within $|\Delta\mathrm{GC}|$ quartiles. Smaller single-sequence gLMs (dashed) hold no edge over composition in any bin; alignment-based and large models keep a stable edge.

neutral rate. Evo2-40B is intermediate (-0.08 to -0.18), Evo2-7B weaker (-0.03 to -0.13), and the smaller single-sequence models (Nucleotide Transformer, Caduceus, HyenaDNA) are near zero. Conservation baselines show similar coupling to the models. The confound is present but is not uniform across the model class.

This is not an artifact of the labelled sets. At genome-wide neutral sites, GPN-MSA's mean score per pentanucleotide context tracks the Carlson neutral mutation rate at $\rho = 0.63$ ($p < 10^{-16}$, 3,072 contexts, 3.6×10^7 scores): highly mutable contexts, above all CpG, receive less deleterious scores (-0.10 versus -1.33 elsewhere). The model has learned the neutral spectrum, the very component that leaks into LLR under non-random ascertainment.

That ascertainment is visible in the labels: ClinVar benign SNVs sit at higher mutation-rate contexts than pathogenic ones (0.033 versus 0.028, $p < 10^{-140}$), and are more often transitions (76% versus 65%) and CpG (35% versus 31%). A score encoding the neutral spectrum thus separates the classes with no regulatory understanding.

4.2. E2: a composition baseline recovers much of aggregate performance

On ClinVar the composition-only classifier reaches AUROC 0.720 (bootstrap 95% confidence interval, CI, 0.716–0.722) with no sequence context. The leading models reach 0.958 (GPN-MSA), 0.960 (Evo2-7B), 0.963 (Evo2-40B), 0.986 (CADD) and 0.912 (phyloP), so the recovery ratio $(\mathrm{comp} - 0.5)/(\mathrm{model} - 0.5)$ is 0.45–0.53 for every one (all DeLong $p \approx 0$). On composition-matched TraitGym the same baseline reaches only 0.586 (Mendelian) and 0.520 (complex), with recovery ratios of 0.18–0.25 for the leading models (GPN-MSA, Evo2-40B, CADD, phyloP); weaker gLMs such as Evo2-7B (0.41) and SpeciesLM (0.72) recover higher because they sit closer to the composition baseline. On GLRB causal eQTL the baseline (0.596) matches or exceeds GPN-MSA (0.560). This roughly two-fold gap between unmatched and matched benchmarks (Table 1, Figure 3) is significant: the composition CIs on ClinVar, Mendelian and complex (0.716–0.722, 0.556–0.613,

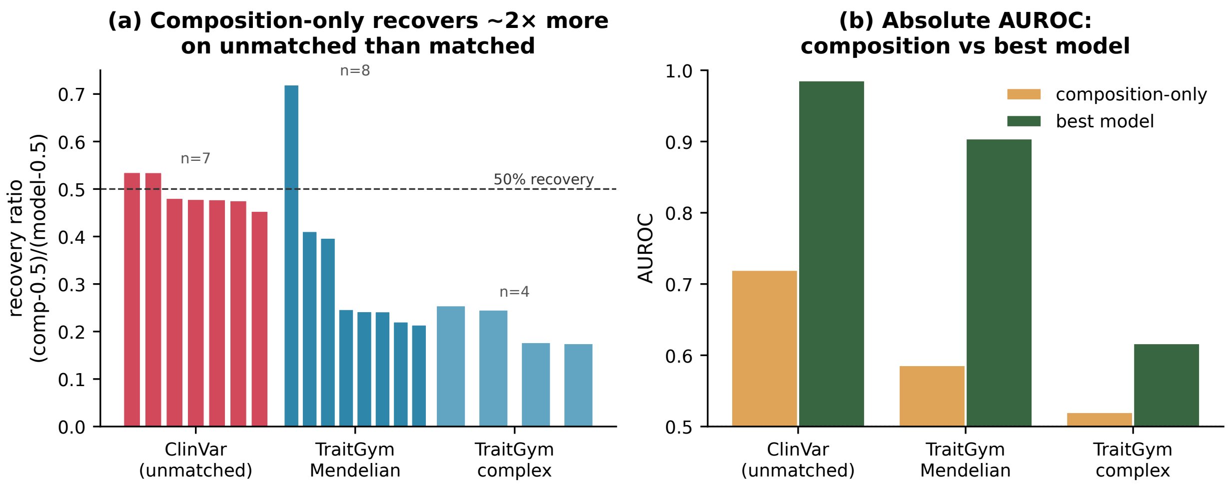

Figure 3: **Composition recovery.** (a) Recovery ratio per model, grouped by benchmark; composition recovers about twice as much above-chance AUROC on unmatched ClinVar as on matched TraitGym. (b) Absolute AUROC of the composition baseline versus the best model per benchmark.

0.503–0.540) do not overlap. Matching removes about half of the confound, a positive result for TraitGym, but a measurable residual leaks past even MAF- and LD-matched controls.

The ClinVar result is not an artifact of coding overlap. Restricting to the strict-noncoding subset that excludes the 46,852 variants annotated as coding (106,656 SNVs remain) leaves the composition baseline at AUROC 0.734 and every recovery ratio in the 0.48–0.50 range, essentially unchanged from the full set.

Nor is it an artifact of ClinVar-benign ascertainment. Swapping the ClinVar-benign controls for gnomAD common variants (allele frequency ≥ 0.05), the field-standard proxy-neutral control, on the same pathogenic set makes the confound stronger: the composition baseline rises to AUROC 0.777 and its recovery ratio against GPN-MSA rises from 0.48 to 0.57, since common variants concentrate at the high-mutation-rate contexts the composition features encode.

Because the recovery ratio is not a decomposition, we also test unique contribution by combining composition and model features under the same folds. On ClinVar, adding GPN-MSA to composition raises AUROC from 0.717 to 0.971 (+0.254), while adding composition to GPN-MSA adds only +0.013 (Evo2-40B: +0.193 versus +0.057). The strong models thus subsume composition and carry large independent signal on top of it, whereas the smaller single-sequence gLMs do not exceed composition (Table 1); the confound inflates the leaderboard *impression* without making the stronger models reducible to composition.

4.3. E3: matching shrinks but does not erase the gLM edge

Figure 2(b) bins TraitGym Mendelian variants by $|\Delta GC|$ and plots each model's AUROC minus the composition AUROC. Nucleotide Transformer, Caduceus and HyenaDNA hold no edge in any bin, so their apparent VEP signal is composition. GPN-MSA, Evo2-40B, CADD and phyloP keep a stable edge of roughly 0.24. We then nearest-neighbour match positives to controls on the C1/C2 features within consequence class, on top of TraitGym's own matching. Model AUROC drops only slightly under this stricter matching (ClinVar GPN-MSA −0.006, TraitGym Mendelian Evo2-

Table 1: Leave-one-chromosome-out AUROC with per-chromosome stratified-bootstrap 95% CIs, AUPRC, and the composition recovery ratio $(\mathrm{AUROC}_{\mathrm{comp}} - 0.5)/(\mathrm{AUROC}_{\mathrm{model}} - 0.5)$ for the composition baseline versus each model. A recovery ratio above one (marked >1) means the model does not exceed the composition baseline. Composition and model AUROC CIs are non-overlapping on every matched-vs-unmatched contrast, and all model-vs-composition DeLong tests give $p < 10^{-3}$ except the near-chance single-sequence gLMs on complex traits ($p > 0.7$).

Method	AUROC [95% CI]	AUPRC	Recovery
ClinVar (unmatched)			
Composition-only	0.720 [0.716,0.723]	0.362	–
GPN-MSA	0.958 [0.957,0.960]	0.861	0.48
Evo2-7B	0.960 [0.959,0.962]	0.882	0.48
Evo2-40B	0.963 [0.962,0.965]	0.870	0.47
CADD	0.986 [0.985,0.987]	0.958	0.45
phyloP	0.912 [0.909,0.914]	0.775	0.53
TraitGym Mendelian (matched)			
Composition-only	0.586 [0.559,0.617]	0.153	–
GPN-MSA	0.904 [0.886,0.921]	0.694	0.21
Evo2-7B	0.710 [0.682,0.743]	0.385	0.41
Evo2-40B	0.851 [0.830,0.872]	0.557	0.25
NT-2.5B	0.534 [0.508,0.566]	0.120	>1
Caduceus	0.509 [0.483,0.539]	0.108	>1
CADD	0.893 [0.871,0.914]	0.712	0.22
phyloP	0.857 [0.830,0.883]	0.655	0.24
TraitGym complex (matched)			
Composition-only	0.520 [0.499,0.537]	0.115	–
GPN-MSA	0.580 [0.562,0.600]	0.212	0.25
Evo2-40B	0.516 [0.498,0.534]	0.137	1.26
NT-2.5B	0.516 [0.500,0.534]	0.114	>1
Caduceus	0.518 [0.502,0.536]	0.114	>1
CADD	0.616 [0.598,0.637]	0.250	0.18
phyloP	0.583 [0.565,0.604]	0.184	0.24
GLRB eQTL			
Composition-only	0.596 [0.585,0.607]	0.633	–
GPN-MSA	0.560 [0.548,0.571]	0.599	>1

40B −0.019, Nucleotide Transformer −0.025), and the model ranking reorders mildly (Kendall $\tau = 0.81$–0.91). The models that clear chance therefore carry composition-independent signal.

4.4. E4: within fixed background the leakage is negligible

The Kircher screens fix the background sequence within each element, which isolates the substitution and its immediate context. For each of the 21 elements we regress GPN-MSA LLR on the measured $\log_2$ effect and compare the raw Spearman correlation with a partial correlation controlling for substitution type, ΔGC and CpG status. The partial and raw correlations nearly coincide (mean absolute change below 0.02; Figure 4(a)), so within a fixed background the model–assay cor-

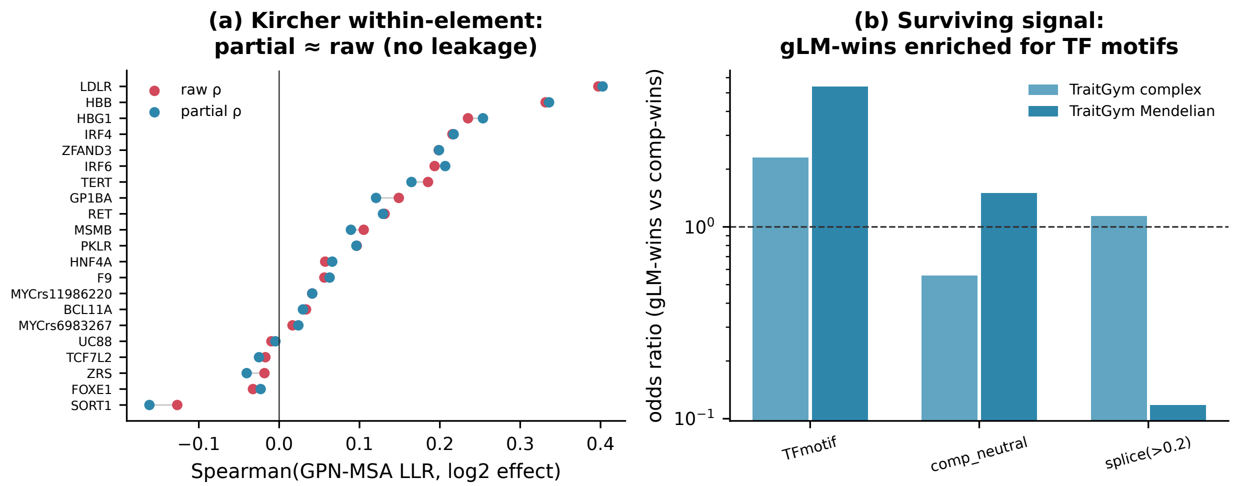

Figure 4: **Localisation and surviving signal.** (a) Kircher within-element raw versus partial Spearman of GPN-MSA LLR against measured effect; the two nearly coincide. (b) Odds ratios for gLM-wins versus composition-wins among positives; gLM-wins are enriched for TF-motif and footprint overlap.

relation is not a composition artifact. Combined with E2, this localises the confound to across-set ascertainment rather than to local context.

4.5. E5: a model-agnostic pentanucleotide calibration

We define a post-hoc transform that subtracts a per-pentanucleotide-context neutral score, estimated leave-one-chromosome-out from control variants only, so that no test-fold label is used. On ClinVar this halves the mutation-rate coupling of GPN-MSA (partial ρ from -0.33 to -0.17) and reduces Evo2-40B coupling, at an AUROC cost below 0.005 (Figure 5). The leaderboard reorders mildly (Kendall $\tau = 0.80$ on ClinVar, 0.91 on TraitGym). A label-free alternative estimator, a genome-wide neutral table built from 3,072 pentanucleotide contexts and 3.6×10^7 sampled genome-wide GPN-MSA scores in the manner of Benegas et al. (2025c), reproduces the effect and is if anything cleaner: it reduces GPN-MSA's mutation-rate coupling from 0.33 to 0.18 while leaving AUROC unchanged (0.958 to 0.961). The small AUROC effect is consistent with E3 and E4: calibration is a decontamination diagnostic that removes the mutation-rate coupling without destroying the signal the stronger models genuinely carry. We release the transform as a drop-in `calibrate_llr` function together with the per-context neutral tables.

4.6. E6: surviving signal is motif-disrupting

Among positives, we compare variants that the calibrated GPN-MSA ranks highly but the composition baseline misses ("gLM-wins") against the reverse ("composition-wins"). gLM-wins are enriched for overlap with transcription-factor (TF) motifs and ChIP footprints, with odds ratio 5.4 on TraitGym Mendelian and 2.3 on complex (Figure 4(b)); the Mendelian estimate rests on a small composition-win set and should be read as directional. Splice-proximity and composition-neutral enrichments are weaker and noisier, as expected given how few splice variants fall in the

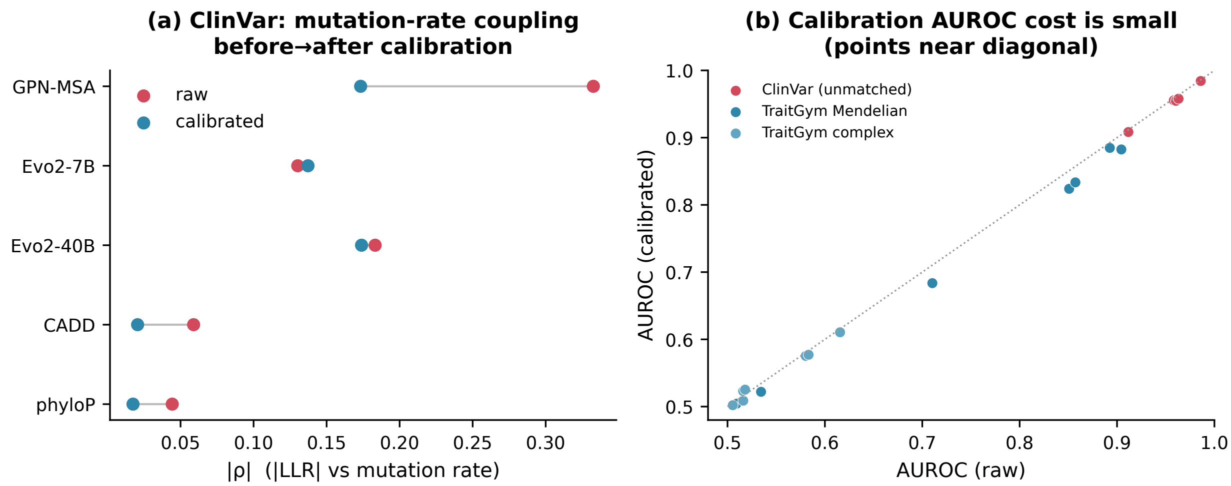

Figure 5: **Post-hoc calibration.** (a) Mutation-rate coupling before and after calibration on ClinVar; the most coupled model (GPN-MSA) is halved. (b) AUROC before versus after calibration; points lie near the diagonal, so the decontamination cost is small.

noncoding-regulatory set. The signal that survives decontamination concentrates on regulatory grammar, namely motif disruption, which is where the gLM claim is genuinely testable.

5. Discussion

Our claim is specific, not a blanket dismissal. A context-free composition and mutation-rate baseline recovers about half of the apparent zero-shot VEP performance of leading gLMs on the unmatched, region-heterogeneous benchmarks that dominate leaderboards, and about a fifth on matched benchmarks. The confound bites hardest on the smaller single-sequence gLMs, which do not exceed the baseline on any benchmark, consistent with Tang et al. (2025). Alignment-based and large autoregressive models keep signal that survives matching and concentrates on motif-disrupting variants. The confound is an across-set ascertainment phenomenon, not a local-context artifact.

Limitations. We score zero-shot LLR, the deployed clinical setting and the headline claim in gLM VEP papers; fine-tuning is out of scope. The recovery ratio depends on classifier capacity, but a supervised one-hot CNN on the same windows stays below both the composition baseline and the gLMs everywhere (AUROC 0.52–0.56), so the baseline is not merely a strong learner. The ClinVar effect is unchanged on the strict noncoding subset (Section 4.2) and strengthens under gnomAD-common controls rather than depending on ClinVar-benign ascertainment. Composition and function are genuinely correlated, so the goal is not to call composition non-biological but to test whether a model advertised as learning regulatory grammar reduces to a mutation-rate detector on the benchmarks used to certify it. In practice, report a composition baseline alongside gLM VEP, prefer matched or calibrated scores for cross-model comparison, and evaluate on motif-disrupting variants.

Reproducibility. All model scores are authors' released per-variant values or public annotations; composition features use the Carlson pentanucleotide table and hg38. Code, the per-context neutral tables, and the released `calibrate_llr` transform are at https://github.com/rikhinkavuru/composition-confound-vep.

References

Anonymous. A mechanistic invariance test reveals compositional shortcuts in genomic language models. *arXiv*, 2026. doi: 10.48550/arXiv.2604.06549. arXiv:2604.06549.

Gonzalo Benegas, Carlos Albors, Alan J Aw, Chengzhong Ye, and Yun S Song. A dna language model based on multispecies alignment predicts the effects of genome-wide variants. *Nature Biotechnology*, 43:1960–1965, 2025a. doi: 10.1038/s41587-024-02511-w.

Gonzalo Benegas, Gökcen Eraslan, and Yun S Song. Benchmarking dna sequence models for causal regulatory variant prediction in human genetics. *bioRxiv*, 2025b. doi: 10.1101/2025.02.11.637758.

Gonzalo Benegas et al. Predicting functional constraints across evolutionary timescales with phylogeny-informed genomic language models. *bioRxiv*, 2025c. doi: 10.1101/2025.09.21.677619. PMC12458161.

Garyk Brixi et al. Genome modeling and design across all domains of life with evo 2. *Nature*, 2026. doi: 10.1038/s41586-026-10176-5.

Jedidiah Carlson, Adam E Locke, et al. Extremely rare variants reveal patterns of germline mutation rate heterogeneity in humans. *Nature Communications*, 9:3753, 2018. doi: 10.1038/s41467-018-05936-5.

ENCODE Project Consortium. Expanded encyclopaedias of dna elements in the human and mouse genomes. *Nature*, 583:699–710, 2020. doi: 10.1038/s41586-020-2493-4.

Gupta, Alan M Moses, and Alex X Lu. Representation learning methods for single-cell microscopy are confounded by background cells. *Proceedings of Machine Learning Research*, 311:162–177, 2025.

InstaDeep et al. Genomics long-range benchmark. In *OpenReview*, 2024. ID Cdc90HKs1I.

Martin Kircher, Chenling Xiong, Beth Martin, et al. Saturation mutagenesis of twenty disease-associated regulatory elements at single base-pair resolution. *Nature Communications*, 10:3583, 2019. doi: 10.1038/s41467-019-11526-w.

Allan Lu, Liu, Lin, and Nadav Brandes. Genomic heterogeneity inflates the performance of variant pathogenicity predictions. *bioRxiv*, 2025. doi: 10.1101/2025.09.05.674459.

Mathur and Ravi Sachidanandam. Benchmarking dna foundation models: Biological blind spots in evo2 variant-effect prediction. *bioRxiv*, 2026. doi: 10.64898/2026.03.10.710786.

Philipp Rentzsch, Daniela Witten, Gregory M Cooper, Jay Shendure, and Martin Kircher. Cadd: predicting the deleteriousness of variants throughout the human genome. *Nucleic Acids Research*, 47:D886–D894, 2019. doi: 10.1093/nar/gky1016.

Rochkoulets, Lovro Vrček, and Mile Šikić. Entropy, disagreement, and the limits of foundation models in genomics. *arXiv*, 2026. arXiv:2604.04287.

Ziqi Tang, Nikunj Somia, Yu, and Peter K Koo. Evaluating the representational power of pre-trained dna language models for regulatory genomics. *Genome Biology*, 2025. doi: 10.1186/s13059-025-03674-8.